

Shape of Data Distributions

Lesson 8-7

Name: _____

Date: _____

Class: _____

Key Vocabulary

Level 1 support

Picture first, then the word, then a plain-language meaning. Say each word out loud.

Heights: 60, 62, 64, 66, 68, 66, 64, 62, 60 — rises and falls evenly like a hill

Symmetric

Data that looks about the same on the left and right.

Skewed right: most data low (1,2,2,3,3,3,15) — tail stretches toward the high value

Skewed

Data bunched on one side with a tail on the other.

Scores 18, 19, 20, 21, 22 form a cluster around 20 — they are tightly grouped

Cluster

A group of numbers that are close together.

Data: 5, 6, 7, 8, ____, ____, ____, 20 — the empty space from 9 to 19 is a gap

Gap

A big empty space where there is no data.

Looking at a dot plot or histogram, you can see if data is symmetric, skewed, clustered, or has gaps

Data distribution

How the data looks: where it sits and how spread out it is.

Symmetric data with low variability: tightly packed around the center. Skewed data: spread unevenly

Variability

How spread out the numbers are.

Key Ideas & Notes

- The league analyst is studying three different data sets from the season: (1) Player heights across all teams, (2) Number of 3-pointers per game by each player, and (3) Years of experience for all coaches.
- Each data set has a different shape when graphed.
- Your job is to describe each shape and explain what it tells us about the data.

Think About It

- Do the three data sets look the same when graphed?
- Which graph has data bunched in the middle? Which has a long tail?
- Are there any gaps or clusters you can spot?

My Notes

Turn & Talk — Discussion Points

Level 1 support

Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

LAUNCH

The analyst graphs player heights, 3-pointers per game, and coaches' years of experience. Why might each data set have a DIFFERENT shape when graphed?

Level 1 support

Start here: Different real-world data sets form different distribution shapes: symmetric, skewed, or clustered.

The shapes are different because the data ____.

Las formas son diferentes porque los datos ____.

A symmetric shape looks ____, while a skewed shape looks ____.

Una forma simétrica se ve ____, mientras que una sesgada se ve ____.

Word bank: symmetric skewed cluster gap distribution

Listen for: Students explain different data produces different shapes and can describe symmetric (balanced) versus skewed (lopsided) distributions.

Level 2

Predict: which of the three data sets is most likely to be skewed, and why?

- I predict ____ is most likely skewed because ____.
- The values would pile up on one side since ____.

EXPLORE

As you match each data description to its shape, how can you tell a symmetric distribution from a skewed one just by looking?

Level 1 support

Start here: A symmetric distribution is balanced around the center; a skewed one has a longer tail on one side.

A symmetric distribution has data that is ____ on both sides.

Una distribución simétrica tiene datos ____ en ambos lados.

A skewed distribution has a longer ____ on one side.

Una distribución sesgada tiene una ____ más larga en un lado.

Word bank: (symmetric) (skewed) (cluster) (gap) (tail)

Listen for: A strong answer says symmetric data mirrors around the center while skewed data has a longer tail pulling to the left or right.

Level 2

Compare: how do clusters and gaps in a distribution give you extra information beyond just 'symmetric' or 'skewed'?

- A cluster tells me ____, while a gap tells me ____.
- These features matter because ____.

CONNECT

Marathon times cluster at 3.5-4.5 hours, with elites at 2.0-2.5 and a long tail of slower runners at 5-7 hours. What is the shape, and should the reporter use the mean or median for a 'typical' time?

Level 1 support

Start here: A skewed distribution is better summarized by the median than the mean.

This distribution is skewed to the ____ because of the long tail of ____.

Esta distribución está sesgada hacia la ____ por la cola larga de ____.

The reporter should use the ____ because ____.

El reportero debería usar la ____ porque ____.

Word bank: (symmetric) (skewed) (cluster) (gap) (distribution)

Listen for: Students identify the right-skewed shape (long tail of slow runners) and choose the median because the slow tail would pull the mean up.

Level 2

Critique: the reporter wants the finish times to sound fast. Which measure might they pick, and would that be honest given the shape?

- To sound fast, the reporter might pick the ____ because ____.
- That would be honest/misleading because the shape is ____.

REFLECT

A data set has most values clustered between 40-60 with a few at 90-100. What is the shape of the distribution, and which measure of center is best?

Level 1 support

Start here: A few high values create a right-skewed shape, so the median is the better center.

The shape is skewed to the ____ because of the values at ____.

La forma está sesgada hacia la ____ por los valores en ____.

The best measure of center is the ____ because ____.

La mejor medida de centro es la ____ porque ____.

Word bank: (symmetric) (skewed) (cluster) (gap) (median)

Listen for: Students describe a right-skewed shape (cluster at 40-60, gap, then 90-100) and choose the median because the high values would inflate the mean.

Level 2

Generalize: state a rule connecting the SHAPE of a distribution to the best measure of center to use.

- If the shape is symmetric, the best center is the ____.
- If the shape is skewed, the best center is the ____ because ____.

Guided Examples

Example 1

A histogram of basketball players' ages shows: most players are 22–28, with a long tail of older players up to age 40. What is the shape of this distribution?

Solution: Most data is on the left (younger ages) with a tail stretching right (older ages). This is skewed right.

Answer: A. Skewed right

Example 2

A data set is symmetric. What is likely true about the mean and median?

Solution: In a symmetric distribution, the mean and median are approximately equal because the data is balanced on both sides.

Answer: A. They are approximately equal

Example 3

Which word describes the direction of the TAIL, not the peak?

Solution: The skew direction is named after the TAIL. If the tail stretches to the right, it is skewed right, even though the peak is on the left.

Answer: A. Skew — the direction of the long tail tells you the skew direction

Write About the Math

The Writing Revolution

I can explain my description using the words symmetric, skewed, cluster, and gap.

1. Kernel Sentence subject + verb

Model: Data distribution is how the data looks: where it sits and how spread out it is.

Distribución de datos es cómo se ven los datos: dónde están y qué tan separados están.

Write a kernel sentence about data distribution. Use a subject and a verb.

Escribe una oración base sobre distribución de datos. Usa un sujeto y un verbo.

2. Sentence Expansion because · but · so

Kernel: Data distribution matters in math

Distribución de datos importa en matemáticas

Expand the kernel three ways. Add a reason, a contrast, and a result.

because
porque

Data distribution matters in math because ____.

Distribución de datos importa en matemáticas porque ____.

but
pero

Data distribution matters in math, but ____.

Distribución de datos importa en matemáticas, pero ____.

so
entonces

Data distribution matters in math, so ____.

Distribución de datos importa en matemáticas, entonces ____.

3. Sentence Types 4 ways to write a math idea

Statement *Afirmación*

Tell one true fact about data distribution.
Di un hecho verdadero sobre data distribution.

Data distribution ____.

Question *Pregunta*

Ask a question about data distribution.
Haz una pregunta sobre data distribution.

How does ____ ?

¿Cómo ____ ?

Exclamation *Exclamación*

Show excitement about data distribution.
Muestra entusiasmo sobre data distribution.

Wow, ____ !

¡Guau, ____ !

Command *Mandato*

Tell a partner what to do with data distribution.
Dile a un compañero qué hacer con data distribution.

First, ____ .

Primero, ____ .

4. Explain Your Reasoning use a sentence starter

The shapes are different because the data ____.

Las formas son diferentes porque los datos ____.

A symmetric shape looks ____, while a skewed shape looks ____.

Una forma simétrica se ve ____, mientras que una sesgada se ve ____.

A symmetric distribution has data that is ____ on both sides.

Una distribución simétrica tiene datos ____ en ambos lados.

| Try It

Solve on your own. Check the answer key when you are done.

1. A dot plot of goals scored per game shows: 0(5 dots), 1(7 dots), 2(4 dots), 3(2 dots), 4(1 dot). What is the shape?

- A. Skewed right — most values at the low end with a tail to the right
- B. Symmetric — equal on both sides
- C. Skewed left — most values at the high end
- D. Uniform — all bars the same height

Show your work:

2. Match the data shape with the best measure of center.

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

A teacher shows two dot plots of quiz scores from two different classes. Class A's data is symmetric with a cluster around 80. Class B's data is skewed right with most scores between 60–70 and a few scores near 100. Compare the classes: which has a higher median? Which has a higher mean? In which class does the mean better represent a typical score?

Sentence starter: Class A is ____ with scores clustered around _____. Class B is skewed ____ with most scores at _____. Class A likely has a ____ median. Class B's mean is pulled ____ by the high scores. The mean better represents a typical score in Class ____ because _____.

Show your work:

Reflect — Exit Ticket

A data set has most values clustered between 40–60, with a few values at 90–100. What is the shape of the distribution and which measure of center is best?

- A. Skewed right; use median
- B. Symmetric; use mean
- C. Skewed left; use median
- D. Uniform; use mean

Your answer:

Answer Key & Teacher Guide

1. **Try It 1:** A. Skewed right — most values at the low end with a tail to the right — *The data peaks at 0–1 goals and tails off toward 4. Most data is on the left with fewer values stretching right — this is skewed right.*
2. **Try It 2:**
3. **Exit Ticket:** A. Skewed right; use median — *The data clusters on the left (40–60) with a tail to the right (90–100). This is skewed right. The median is best because the mean would be pulled toward the high values.*

Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Data distribution is how the data looks: where it sits and how spread out it is.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).